KISHINCHAND CHELLARAM COLLEGE, MUMBAI - 20.

M. Sc. (Computer Science) Year_____.

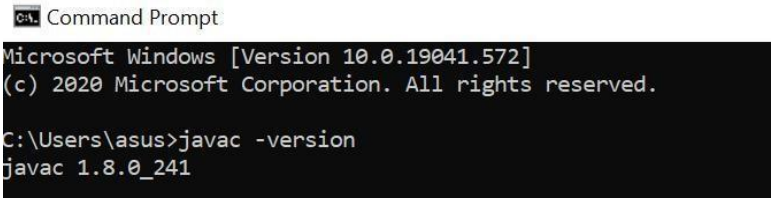
Practical 01: HADOOP INSTALLATIONS

Steps to Install Hadoop

- Install Java JDK 1.8
- Download Hadoop and extract and place under C drive
- Set Path in Environment Variables
- Config files under Hadoop directory
- Create folder datanode and namenode under data directory
- Edit HDFS and YARN files
- Set Java Home environment in Hadoop environment
- Setup Complete. Test by executing start-all.cmd

1. **Install Java**

https://www.oracle.com/java/technologies/javase-jdk8-downloads.html

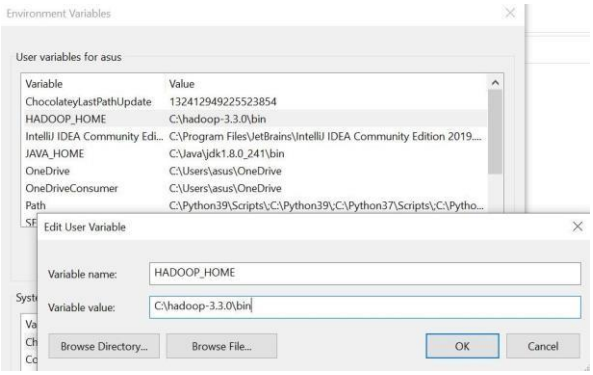


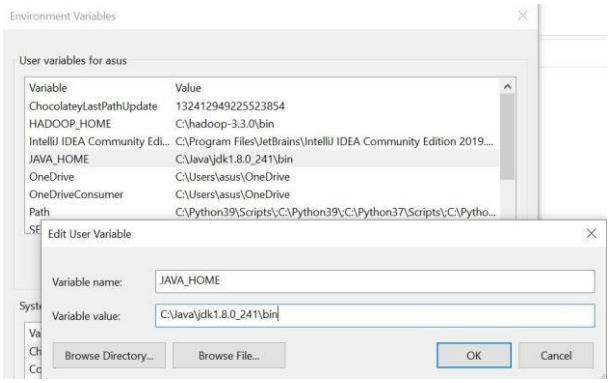
2. **Download Hadoop**

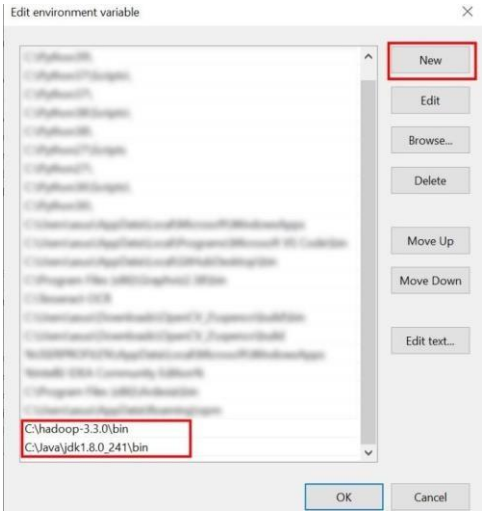
   – **https://www.apache.org/dyn/closer.cgi/hadoop/common/hadoop-3.3.0/hadoop3.3.0.tar.gz**

   – **extract to C:\Hadoop**

3. **Set the path JAVA_HOME Environment variable**
4. **Set the path HADOOP_HOME Environment variable**







5. Configurations

Edit file C:/Hadoop-3.3.0/etc/hadoop/core-site.xml, paste the xml code in folder and save

```
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>
```

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Rename "mapred-site.xml.template" to "mapred-site.xml" and edit this file C:/Hadoop-3.3.0/etc/hadoop/mapred-site.xml, paste xml code and save this file.

```
<configuration>
  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>
</configuration>
```

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Create folder "data" under "C:\Hadoop-3.3.0"

Create folder "datanode" under "C:\Hadoop-3.3.0\data" Create folder "namenode" under "C:\Hadoop-3.3.0\data"

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Edit file C:\Hadoop-3.3.0/etc/hadoop/hdfs-site.xml, paste xml code and save this file.

```
<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
  <property>
    <name>dfs.namenode.name.dir</name>
    <value>/hadoop-3.3.0/data/namenode</value>
  </property>
  <property>
    <name>dfs.datanode.data.dir</name>
    <value>/hadoop-3.3.0/data/datanode</value>
  </property>
</configuration>
```

==========

Edit file C:/Hadoop-3.3.0/etc/hadoop/yarn-site.xml, paste xml code and save this file.

```
<configuration>
```

```
  <property>
        <name>yarn.nodemanager.aux-services</name>
        <value>mapreduce_shuffle</value>
  </property>
  <property>
      <name>yarn.nodemanager.auxservices.mapreduce.shuffle.class</name>
          <value>org.apache.hadoop.mapred.ShuffleHandler</value>
  </property>
</configuration>
```

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Edit file C:/Hadoop-3.3.0/etc/hadoop/hadoop-env.cmd by closing

the command line

"JAVA_HOME=%JAVA_HOME%" instead of set "JAVA_HOME=C:\Java"

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6. **Hadoop Configurations Download**

https://github.com/brainmentorspvtltd/BigData_RDE/blob/master/Hadoop%20Configuration.zip or (for hadoop 3)

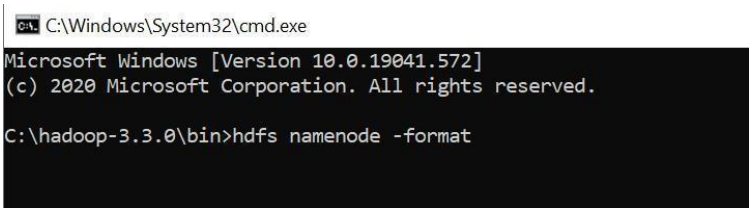
https://github.com/s911415/apache-hadoop-3.1.0-winutils

– Copy folder bin and replace existing bin folder in

C:\Hadoop-3.3.0\bin

– Format the NameNode

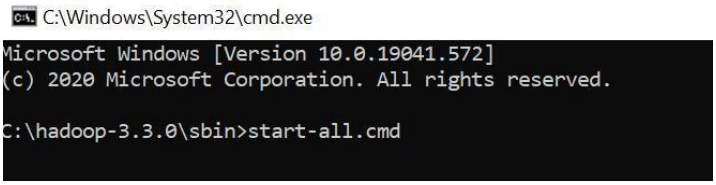
– Open cmd and type command "hdfs namenode –format"



7. Testing

– Open cmd and change directory to C:\Hadoop-3.3.0\sbin

– type start-all.cmd



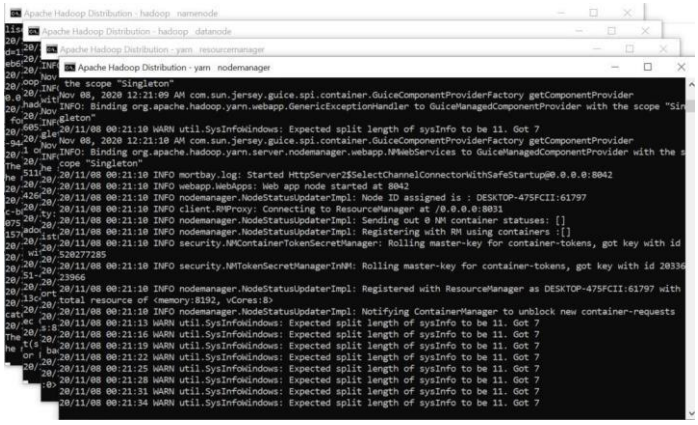
(Or you can start like this)

– Start namenode and datanode with this command

– type start-dfs.cmd

– Start yarn through this command

– type start-yarn.cmd



Make sure these apps are running

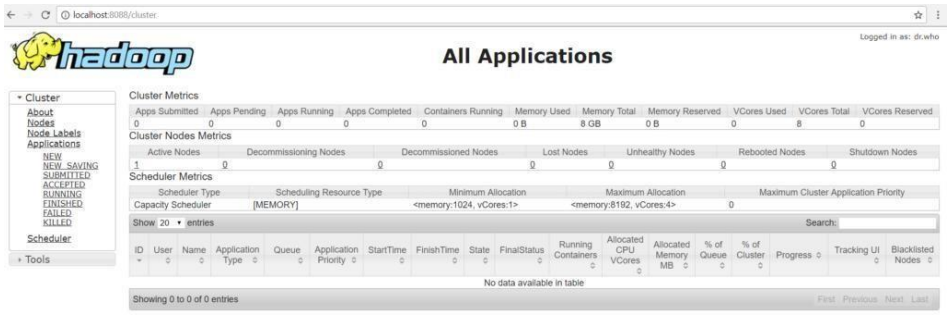
– Hadoop Namenode

– Hadoop datanode

– YARN Resource Manager

– YARN Node Manager

Open: http://localhost:8088



Open: http://localhost:9870

Overview 'localhost:9000' (✔active)

Started:	Sun Nov 08 16:53:46 +0530 2020
Version:	3.3.0,[redacted]9af
Compiled:	Tue Jul 07 00:14:00 +0530 2020 by brahma from branch-3.3.0
Cluster ID:	C[redacted]
Block Pool ID:	B[redacted]44

Summary